HRISHIKESH BELAGALI

Undergraduate Research Assistant

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SUMMARY

PhD applicant in Computer Science with interests in quantum algorithms, molecular dynamics, computational physics, and machine learning.

SKILLS

Languages: Python, Rust, C++, LaTeX

Packages: NumPy, PyTorch, JAX, scipy, Qiskit, Cirq, ruff, mypy, pytest

Softwares: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S) — **Michigan State University Honors College**
Expected Graduation: 2027
Current GPA: 4.0

RELEVANT COURSEWORK

- Graduate Quantum Algorithms
- Graduate Computational Foundations of Artificial Intelligence
- Graduate Tensor Networks (Scheduled)
- Linear Algebra

EXPERIENCE

2026–

Undergraduate Research Assistant — **MSU-Q**
- Conducted research on fermionic backpropagation, a method for performing hybrid Schrödinger-Heisenberg simulation of large-scale unitary cluster Jastrow circuits.
- Implemented fermionic backpropagation into *ffsim*, a Python package for simulating fermionic quantum systems.
- Primary developer of *propaq*, a Python package for Heisenberg propagation, designed for large-scale simulation on HPC clusters.

2025–2026

Honors Research Scholar — **MSU Honors College/MSU-Q**
- Conducted research on algorithms for finding optimal compilation subgroups under the guidance of Professor Ryan LaRose.
- Investigated computational group theory methods for membership in unitary subgroups.
- Implemented compilation optimization for generalized subgroups using Python and Cirq.

2024–

Undergraduate Research Assistant — **Vermaas Lab, Plant Research Laboratory**
- Simulated the effects of small molecules like butanol on *Clostridium acetobutylicum* and *Zymomonas mobilis* membrane dynamics.
- Investigated the interaction of the chloroplast targeting protein (At4g16060) with thylakoid membranes using molecular dynamics.
- Analyzed the effect of PDLP-5 (Plasmodesmata Located Protein) on water flow rates through PIP1;4 aquaporin tetramer using grid-steered molecular dynamics.

PUBLICATIONS

[1] Hrishikesh Belagali, Ryan LaRose. "propaq – A Python package for Heisenberg propagation", *arXiv* 2026, manuscript submitted to *Quantum* (pending review). Available:

[2] Hrishikesh Belagali, Thomas Van Camp, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose, "Efficient classical simulation of large-scale unitary cluster Jastrow circuits," *arXiv*, 2026, manuscript submitted to *Physical Review Letters* (pending review). Available: https://arxiv.org/abs/2607.21337

[3] Hrishikesh Belagali, Nitin Kumar Singh, Josh V. Vermaas, "The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*," *ChemRxiv*, 2026, manuscript submitted to *ACS JPCB* (pending review). Available: https://chemrxiv.org/doi/full/10.26434/chemrxiv.15004463/v1

CONFERENCES

2026
ACS, IL

Molecular mechanisms of alcohol tolerance in Clostridium acetobutylicum and Zymomonas mobilis: The role of cyclopropane fatty acids and hopanoids — **acs.digitellinc.com**
Poster presented at ACS Fall 2026 Conference

2026
MQC, IN

Efficient classical simulation of large-scale unitary cluster Jastrow circuits
Poster presented at MQC Entanglement 2026

2026
GLBRC, WI

The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*
Poster presented at Great Lakes Bioenergy Research Center Annual Science Meeting.

2026
MSU UURAF

Reducing T Gate Counts via Gateset Rotation
Poster presented at MSU Undergraduate Research and Arts Forum. Awarded first place in the Computational Mathematics, Science and Engineering division.

OPEN SOURCE CONTRIBUTIONS

propaq
Python, Rust

propaq **github.com**
Primary developer of *propaq*, a Python package for Heisenberg propagation, designed for large-scale simulation on HPC clusters.
- Implmented thread-safe kernels and compact representation of Pauli and Majorana strings in Rust for optimal multi-threaded simulation.
- Developed a Python interface integrating *propaq* with Qiskit and Cirq for seamless simulation workflows using pyo3 bindings.
- Designed a plugin system for developing custom noise and truncation policies in Python, Rust, C, and Julia to enable user customization with minimal overhead.

ffsim
Python, Rust

ffsim **github.com**
Contributed to *ffsim*, a Python package for simulating fermionic quantum systems.
- Implemented fermionic backpropagation, from "Efficient classical simulation of large-scale unitary cluster Jastrow circuits" (arXiv:2607.21337) into *ffsim* for fast unitary cluster Jastrow ansatz simulation.
- Created a JAX-based implementation for GPU-accelerated variational optimization.

QuTiP
Python

Quantum Toolbox in Python (QuTiP) **github.com**
Contributed to QuTiP, an open-source software for simulating quantum systems.
- Implemented functionality to return eigenkets as an operator in the Qobj class.

Python
NumPy

aqua-blue **github.com/pypi.org**
Secondary maintainer of aqua-blue, a package building echo state networks.
- Implemented reservoir state time-lag, sparsity and spectral radius to improve prediction accuracy.
- Introduced timezone-aware datetime support for seamless interfacing with data pipelines.
- Demonstrated prediction capability by predicting quantum time evolution of Gaussian wave packets.

Python
NumPy
hyperopt

aqua-blue-hyperopt **github.com/pypi.org**
Primary maintainer of aqua-blue-hyperopt, a package implementing hyperparameter optimization functionality for aqua-blue models.
- Created modular hyperparameter optimization functionality for aqua-blue reservoir computers.
- Implemented custom loss function capabilities for flexible model evaluation.

tce-lib
Python

tce-lib **github.com**
Contributed to tce-lib, a Python package for building tensor cluster expansion models for concentrated alloys.
- Implemented lazy evaluation of Monte Carlo trajectories for efficient simulation on HPC clusters.

HONORS AND AWARDS

- Wielenga Research Scholarship, Honors College (2025-2026)
- Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)

PROJECTS

Python
Qiskit

Adiabatic Quantum Computation **github.com**
Implemented a quantum adiabatic algorithm to solve QUBO type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques.

Python
OVITO
pyMOL

Genetic Annealing to Determine Protein Structures **github.com**
Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irbäck's off-lattice model energy equation (RMSD < 3.0).